

Question 5:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc form.c -o form
himasree@LAPTOP-41COT5AA:~/OSLab$ ./form
Parent Process
Child Process
PID : 938
Child PID : 939
PID : 939
Parent PID : 938
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
    }
    else
    {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }

    return 0;
}
```

Question 6:

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec()\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This line will not execute if exec succeeds.\n");

    return 0;
}
```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano excel.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc excel.c -o excel
himasree@LAPTOP-41COT5AA:~/OSLab$ ./excel
Before exec()
total 372
drwxr-xr-x 3 himasree himasree 4096 Aug  1 03:05 MYProject
drwxr-xr-x 7 himasree himasree 4096 Aug  1 03:33 MyProject
-rw-r--r-- 1 himasree himasree  731 Aug  5 03:23 copy.c
-rw-r--r-- 1 himasree himasree   33 Aug  5 03:30 copy.txt
-rwxr-xr-x 1 himasree himasree 16000 Aug  5 03:55 excel
-rw-r--r-- 1 himasree himasree  205 Aug  5 03:55 excel.c
-rw----- 1 himasree himasree  205 Aug  5 03:52 excel.c.save
drwxr-xr-x 2 himasree himasree 4096 Jul 31 09:22 exec
-rwxr-xr-x 1 himasree himasree 16008 Jul 30 09:58 exp1
-rw-r--r-- 1 himasree himasree   99 Jul 29 03:04 exp1.c
-rwxr-xr-x 1 himasree himasree 16000 Jul 30 09:59 exp2
-rw-r--r-- 1 himasree himasree   94 Jul 29 03:06 exp2.c
-rwxr-xr-x 1 himasree himasree 16000 Jul 30 09:59 exp3
-rw-r--r-- 1 himasree himasree  163 Jul 29 03:07 exp3.c
-rwxr-xr-x 1 himasree himasree 16088 Jul 30 09:59 exp4
-rw-r--r-- 1 himasree himasree  278 Jul 29 03:08 exp4.c
-rwxr-xr-x 1 himasree himasree 16040 Jul 30 09:59 exp5
-rw-r--r-- 1 himasree himasree  214 Jul 29 03:10 exp5.c
-rwxr-xr-x 1 himasree himasree 16048 Jul 30 09:59 exp6
-rw-r--r-- 1 himasree himasree  116 Jul 29 03:12 exp6.c
-rwxr-xr-x 1 himasree himasree 16040 Jul 30 09:59 exp7
-rw-r--r-- 1 himasree himasree  220 Jul 29 03:14 exp7.c

```

Question 7.a:

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano q3.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc q3.c -o q3
himasree@LAPTOP-41COT5AA:~/OSLab$ ./q3
Parent Waiting...
Child Process Running
Child Process Finished
Parent Continues After Child Finishes

```

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process Running\n");
        sleep(3);
        printf("Child Process Finished\n");
    }
    else
    {
        printf("Parent Waiting...\n");
        wait(NULL);
        printf("Parent Continues After Child Finishes\n");
    }

    return 0;
}

```

Question 7.b:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano q4.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc q4.c -o q4
himasree@LAPTOP-41COT5AA:~/OSLab$ ./q4
Child Process
Parent resumes after child.
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        sleep(2);
    }
    else
    {
        waitpid(pid, NULL, 0);
        printf("Parent resumes after child.\n");
    }

    return 0;
}
```

Question 8:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc q5.c -o q5
himasree@LAPTOP-41COT5AA:~/OSLab$ ./q5
PID = 1001, PPID = 944
PID = 1003, PPID = 1001
himasree@LAPTOP-41COT5AA:~/OSLab$ PID = 1002, PPID = 943
PID = 1004, PPID = 1002
pstree
systemd--agetty
          |--cron
          |--dbus-daemon
          |--init-systemd(Ub)
          |   |--SessionLeader--Relay(944)--bash--pstree
          |   |--init--{init}
          |   |--login--bash
          |   |--{init-systemd(Ub)}
          |--polkitd--3*[{polkitd}]
          |--rsyslogd--3*[{rsyslogd}]
          |--systemd--(sd-pam)
          |--systemd-journal
          |--systemd-logind
          |--systemd-resolve
          |--systemd-timesyn--{systemd-timesyn}
          |--systemd-udev--25*[{udev-worker}]
          |--unattended-upgr--{unattended-upgr}
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    fork();

    printf("PID = %d, PPID = %d\n", getpid(), getppid());

    return 0;
}
```

Question 9:

```

$ nmapree@LAPTOP-41COTSA5A:~/OSLab$ strace ls
execve("/usr/bin/lis", ["lis"], 0x7ffe4724aa60 /* 26 vars */) = 0
brk(NULL) = 0x63cbc8d7f000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7c274e310000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25767, ...}) = 0
mmap(NULL, 25767, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7c274e309000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\220\243\2\0\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c274e2dc000
mmap(0x7c274e2e2000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x7c274e2e2000
mmap(0x7c274e2ff000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x7c274e2ff000
mmap(0x7c274e305000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x7c274e305000

mmap(0x7c274e307000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7c274e307000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\220\243\2\0\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0..." , 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0..." , 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c274e000000
mmap(0x7c274e028000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7c274e028000
mmap(0x7c274e1b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7c274e1b0000
mmap(0x7c274e1ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7c274e1ff000
mmap(0x7c274e205000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7c274e205000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libcpre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0..." , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c274e242000
mmap(0x7c274e244000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7c274e244000
mmap(0x7c274e2b2000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x7c274e2b2000
mmap(0x7c274e2da000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x7c274e2da000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7c274e23f000
arch_prctl(ARCH_SET_FS, 0x7c274e23fad0) = 0
set_tid_address(0x7c274e23fad0) = 1010
set_robust_list(0x7c274e23fae0, 24) = 0
rseq(0x7c274e240120, 0x20, 0, 0x53053053) = 0
mprotect(0x7c274e1ff000, 16384, PROT_READ) = 0
mprotect(0x7c274e2da000, 4096, PROT_READ) = 0
mprotect(0x7c274e305000, 4096, PROT_READ) = 0
mprotect(0x63cbae104000, 8192, PROT_READ) = 0

```

```

openat(AT_FDCWD, "/", O_RDONLY|O_CLOEXEC, 0) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=360460, ...}) = 0
mmap(NULL, 360460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7c274dfa7000
close(3) = 0
ioctl(1, TCGETS, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=IS
IG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=30, ws_col=120, ws_xpixel=0, ws_ypixel=0}) = 0
openat(AT_FDCWD, ".", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
getdents64(3, 0x63cbc8d888a0 /* 49 entries */, 32768) = 1400
getdents64(3, 0x63cbc8d888a0 /* 0 entries */, 32768) = 0
close(3) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
write(1, "MYProject excel\t exp1\t exp3\t ex"..., 91MYProject excel    exp1    exp3    exp5    exp7    exp9    program
    program2    program4    q3    q5
) = 91
write(1, "MyProject excel.c\t exp1.c exp3"..., 107MyProject excel.c exp1.c exp3.c exp5.c exp7.c exp9.c program
.c program2.c program4.c q3.c q5.c
) = 107
write(1, "copy.c\t excel.c.save exp2\t ex"..., 101copy.c    excel.c.save exp2    exp4    exp6    exp8    form
program1    program3    q2    q4    sample.txt
) = 101
write(1, "copy.txt exec\t\t exp2.c exp4.c"..., 97copy.txt    exec    exp2.c exp4.c exp6.c exp8.c form.c
program1.c program3.c q2.c    q4.c
) = 97
close(1) = 0
close(2) = 0
exit_group(0) = ?
*** exited with 0 ***

```

Question 10:

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano shell.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc shell.c -o shell
himasree@LAPTOP-41COT5AA:~/OSLab$ ./shell
MiniShell> ls
MYProject excel.c    exp2    exp4.c exp7    exp9.c    program1    program3.c q3    q5.c
MyProject excel.c.save exp2.c exp5    exp7.c form    program1.c program4    q3.c sample.txt
copy.c    exec    exp3    exp5.c exp8    form.c    program2    program4.c q4    shell
copy.txt exp1    exp3.c exp6    exp8.c program    program2.c q2    q4.c shell.c
excel    exp1.c exp4    exp6.c exp9    program.c program3    q2.c    q5    trace.txt
MiniShell> pwd
/home/himasree/OSLab
MiniShell> date
Wed Aug 5 04:04:43 UTC 2026
MiniShell> exit

```

```

#include <string.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("MiniShell> ");
        scanf("%s", command);

        if(strcmp(command, "exit") == 0)
            break;

        pid_t pid = fork();

        if(pid == 0)
        {
            execlp(command, command, NULL);
            printf("Invalid Command\n");
            exit(1);
        }
    }
}

```

```
        exit(1);
    }
    else
    {
        wait(NULL);
    }
}

return 0;
|
```